Charged Particle in a Magnetic Field: Circular Motion

Charged Particle in a Magnetic Field · oPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://ophysics.com/em7.html
Curriculum links: magnetic fields

Learning intentions

- I can apply F = qvB to explain why a charged particle moves in a circle in a uniform magnetic field.
- I can use r = mv/(qB) to calculate or predict the radius of that circular path.

Getting started

Open the simulation and set the particle's mass, charge, initial velocity and the magnetic field strength. Run the simulation and observe the circular path.

Tasks

1. Before running, use the right-hand (or left-hand, for a negative charge) rule to predict which way the particle will curve. Check your prediction.

2. Using your chosen values, calculate the expected radius using r = mv/(qB), then compare it to the simulation's path.

Working:

3. Keeping mass, charge and velocity constant, run the simulation at four different magnetic field strengths and record the radius each time.

Field strength	Radius

4. Using your table from Task 3, describe the relationship between magnetic field strength and radius. Is it consistent with r = mv/(qB)?

Extension (optional)

This circular motion is the basis of a mass spectrometer, which identifies particles by the radius of their curved path. Explain how measuring the radius, given a known charge, velocity and field, could reveal an unknown particle's mass.